

THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

B. MILLER

ABSTRACT. We give a classical proof of a generalization of the $\mathbb{G}_0$ dichotomy to uniformly analytic families of digraphs indexed by elements of $\kappa^{\mathbb{N}}$ and obtain Kanovei's generalization of the $\mathbb{G}_0$ dichotomy to κ -Souslin digraphs as a corollary.

INTRODUCTION

Suppose that κ is an infinite cardinal and X is a Hausdorff space. Endow κ with the discrete topology and $\kappa^{\mathbb{N}}$ with the product topology. For all $n \in \mathbb{N}$ and $s \in \kappa^n$, set $|s| = n$. For all $\sigma \in \kappa^{\leq \mathbb{N}}$, let $s \sqsubseteq \sigma$ indicate that $\forall i < |s| \ s(i) = \sigma(i)$. Define $\mathcal{N}_s = \{\sigma \in \kappa^{\mathbb{N}} \mid s \sqsubseteq \sigma\}$. We write $\forall^* \sigma \in \kappa^{\mathbb{N}} \ \phi(\sigma)$ to indicate that $\{\sigma \in \kappa^{\mathbb{N}} \mid \phi(\sigma)\}$ is comeager. The *projection* function from $X \times Y$ to X is given by $\text{proj}_X(x, y) = x$. The *horizontal and vertical sections* of a set $R \subseteq X \times Y$ are given by $R^y = \{x \in X \mid x R y\}$ and $R_x = \{y \in Y \mid x R y\}$. Given a function $\pi: X \times Y \rightarrow Z$, define $\pi_x: Y \rightarrow Z$ by $\pi_x(y) = \pi(x, y)$.

We say that $\phi: C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \rightarrow \kappa^{\mathbb{N}} \times X$ is *parameter preserving* if $\text{proj}_{\kappa^{\mathbb{N}}} \circ \phi = \text{proj}_{\kappa^{\mathbb{N}}} \upharpoonright C$, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly analytic (vertical) sections* if there are a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$.

The *diagonal union* of a sequence $(X_s)_{s \in \kappa^{< \mathbb{N}}}$ of subsets of $\kappa^{\mathbb{N}} \times X$ is given by $\bigvee_{s \in \kappa^{< \mathbb{N}}} X_s = \bigcup_{s \in \kappa^{< \mathbb{N}}} X_s \cap (\mathcal{N}_s \times X)$. Countable unions are expressible as diagonal unions, and the vertical sections of diagonal unions are themselves countable unions. We say that a set $B \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly Borel (vertical) sections* if it is in the smallest family of subsets of $\kappa^{\mathbb{N}} \times X$ containing the products of $\kappa^{\mathbb{N}}$ with open subsets of X and closed under complements and diagonal unions. We say that a function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ is *uniformly Borel* if $c^{-1}(\{n\})$ has uniformly Borel sections for all $n \in \mathbb{N}$.

A digraph on X is an irreflexive set $G \subseteq X \times X$. A *homomorphism* from a digraph G on X to a digraph H on Y is a function $\pi: X \rightarrow Y$ such that $(\pi \times \pi)(G) \subseteq H$. A *Y -coloring* of G is a homomorphism $c: X \rightarrow Y$ from G to inequality. Fix $\mathfrak{s} \in \prod_{n \in \mathbb{N}} 2^n$ such that $\mathfrak{s}(\mathbb{N})$ is dense in $2^{< \mathbb{N}}$ and define $\mathbb{G}_0 = \bigcup_{n \in \mathbb{N}} \{(\mathfrak{s}_n \frown (i) \frown c)_{i < 2} \mid c \in 2^{\mathbb{N}}\}$.

Here we establish the following version of the $\mathbb{G}_0$ dichotomy:

Theorem 1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ has uniformly analytic sections that are digraphs on X . Then exactly one of the following holds:*

- (1) *There is a uniformly Borel function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ such that c_σ is a coloring of G_σ for comeagerly many $\sigma \in \kappa^{\mathbb{N}}$.*
- (2) *There exist $t \in \kappa^{<\mathbb{N}}$, a dense G_δ set $H \subseteq \mathcal{N}_t$, and a continuous function $\pi: H \times 2^{\mathbb{N}} \rightarrow X$ such that π_σ is a homomorphism from $\mathbb{G}_0$ to G_σ for all $\sigma \in H$.*

In §1, we note the basic closure properties of sets with uniformly analytic and Borel sections. In §2, we consider how these notions interact with Baire category. In §3, we establish a uniform version of Lusin's separation theorem. In §4, we prove Theorem 1. And in §5, we obtain Kanovei's κ -Souslin generalization [Kan97] of the $\mathbb{G}_0$ dichotomy.

We leave it to the reader to check that these results can be combined with the arguments establishing the usual consequences of the $\mathbb{G}_0$ dichotomy to obtain analogous uniform and κ -Souslin generalizations.

Under $\text{ZF} + \text{DC}$, the arguments can be pushed through by retaining codes for the relevant objects and replacing the choices of representations and separating sets by canonical operations on these codes. We avoid this approach, however, in favor of a clearer presentation.

Under $\text{ZF} + \text{DC} + \text{Det}(\Delta_{2n}^1)$, a more careful version of this book-keeping likely yields the generalization of the $\mathbb{G}_0$ dichotomy to Σ_{2n+1}^1 digraphs. The rough idea would be to use higher sharps and the associated companion theorems to obtain canonical tree representations and parameter-dependent admissible sets whose Σ_1 -definable subsets of $\mathbb{N}^{\mathbb{N}}$ have the required complexity. Carrying out the coded version of the arguments inside these admissible sets should then yield Δ_{2n+1}^1 prewellorderings whose equivalence classes are G -independent.

1. POINTCLASSES

We direct the reader to [Kec95, §14] for the most basic facts concerning analytic sets. The following closure properties can be proven in the same fashion as the corresponding facts for analytic sets:

Proposition 1.1. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under countable intersections and countable unions.*

Proposition 1.2. *Suppose that κ is an aleph, X and Y are Hausdorff spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly analytic sections,*

and $f: A \rightarrow \kappa^{\mathbb{N}} \times Y$ is continuous and parameter preserving. Then $f(A)$ and $f^{-1}(B)$ have uniformly analytic sections.

Given a set $S \subseteq \kappa^{<\mathbb{N}}$ and a sequence $(X_s)_{s \in S}$ of subsets of $\kappa^{\mathbb{N}} \times X$, define $\nabla_{s \in S} X_s = \nabla_{s \in \kappa^{<\mathbb{N}}} X_s$, where $X_s = \emptyset$ for all $s \in \sim S$.

Proposition 1.3. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under diagonal unions.*

Proof. Suppose that $(A_s)_{s \in \kappa^{<\mathbb{N}}}$ is a sequence of subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections and $A = \nabla_{s \in \kappa^{<\mathbb{N}}} A_s$. Then there are closed sets $C_s \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_s: C_s \rightarrow \kappa^{\mathbb{N}} \times X$ with $A_s = \phi_s(C_s)$ for all $s \in \kappa^{<\mathbb{N}}$. As the set $C_n = \nabla_{s \in \kappa^n} C_s$ is closed and the function $\phi_n = \bigcup_{s \in \kappa^n} \phi_s \upharpoonright (C_s \cap (\mathcal{N}_s \times \mathbb{N}^{\mathbb{N}}))$ is continuous, the set $A_n = \nabla_{s \in \kappa^n} A_s = \phi_n(C_n)$ has uniformly analytic sections for all $n \in \mathbb{N}$, as does $A = \bigcup_{n \in \mathbb{N}} A_n$ by Proposition 1.1. $\square$

Proposition 1.4. *Suppose that κ is an aleph and X is a topological space. Then every Borel subset of $\kappa^{\mathbb{N}} \times X$ has uniformly Borel sections.*

Proof. It is sufficient to show that every open set $U \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. For all $s \in \kappa^{<\mathbb{N}}$, let $U_s \subseteq X$ be the maximal open set for which $\mathcal{N}_s \times U_s \subseteq U$. Then $U = \nabla_{s \in \kappa^{<\mathbb{N}}} \kappa^{\mathbb{N}} \times U_s$. $\square$

Proposition 1.5. *Suppose that κ is an aleph, X and Y are topological spaces, $B \subseteq \kappa^{\mathbb{N}} \times Y$ has uniformly Borel sections, and $f: \kappa^{\mathbb{N}} \times X \rightarrow \kappa^{\mathbb{N}} \times Y$ is Borel and parameter preserving. Then $f^{-1}(B)$ has uniformly Borel sections.*

Proof. The family of sets for which the proposition holds contains the open sets by Proposition 1.4 and is clearly closed under complements. The fact that f is parameter preserving ensures that it is also closed under diagonal unions. $\square$

Proposition 1.6. *Suppose that X is a Hausdorff space, $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then $A \cap B$ and $A \setminus B$ have uniformly analytic sections.*

Proof. By the usual proof of the corresponding fact for analytic and Borel sets, we need only note that if $B = \nabla_{s \in \kappa^{<\mathbb{N}}} B_s$ and both $A \cap B_s$ and $A \setminus B_s$ have uniformly analytic sections for all $s \in \kappa^{<\mathbb{N}}$, then $A \cap B = \nabla_{s \in \kappa^{<\mathbb{N}}} A \cap B_s$ has uniformly analytic sections by Proposition 1.3, and $A \setminus B = \bigcap_{n \in \mathbb{N}} \nabla_{s \in \kappa^n} A \setminus B_s$ has uniformly analytic sections by Propositions 1.1 and 1.3. $\square$

2. BAIRE CATEGORY

We direct the reader to [Kec95, §8] for the most basic facts concerning Baire category (such as localization). Although $\kappa^{\mathbb{N}}$ need not be separable, the usual proofs use only the Baire category theorem and go through unchanged for completely metrizable spaces.

Proposition 2.1. *Suppose that κ is an aleph, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then every horizontal section of B has the Baire property.*

Proof. It is enough to show that if $n \in \mathbb{N}$ and the horizontal sections of $B_s \subseteq \kappa^{\mathbb{N}} \times X$ have the Baire property for all $s \in \kappa^{<\mathbb{N}}$, then the horizontal sections of $B = \bigcap_{s \in \kappa^n} B_s$ also have the Baire property. Given $x \in X$, fix an open set $U_s \subseteq \kappa^{\mathbb{N}}$ and dense open sets $U_{m,s} \subseteq \kappa^{\mathbb{N}}$ such that $(B_s^x \triangle U_s) \cap \bigcap_{m \in \mathbb{N}} U_{m,s} = \emptyset$ for all $s \in \kappa^n$. Then $U = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_s$ is open, $U_m = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_{m,s}$ is dense and open for all $m \in \mathbb{N}$, and $(B^x \triangle U) \cap \bigcap_{m \in \mathbb{N}} U_m = \emptyset$, so B^x has the Baire property. $\square$

Proposition 2.2. *Suppose that κ is an aleph, X is a topological space, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections. Then $\text{proj}_{\kappa^{\mathbb{N}}}(A)$ has the Baire property.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$. Then Nikodým's theorem (see [Kec95, Corollary 29.14]) ensures that

$$\text{proj}_{\kappa^{\mathbb{N}}}(A) = \text{proj}_{\kappa^{\mathbb{N}}}(C) = \bigcup_{c \in \mathbb{N}^{\mathbb{N}}} \bigcap_{n \in \mathbb{N}} \overline{\text{proj}_{\kappa^{\mathbb{N}}}(C \cap (\kappa^{\mathbb{N}} \times \mathcal{N}_{c|n}))}$$

has the Baire property. $\square$

3. SEPARATION

A set $B \subseteq X$ separates sets $A_0, A_1 \subseteq X$ if $A_0 \subseteq B$ and $A_1 \cap B = \emptyset$.

Proposition 3.1. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_0, A_1 \subseteq \kappa^{\mathbb{N}} \times X$ are disjoint sets with uniformly analytic sections. Then they are separated by a set with uniformly Borel sections.*

Proof. Fix closed sets $C_i \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_i: C_i \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A_i = \phi_i(C_i)$, and define $A_{i,s,t} = \phi_i(C_i \cap (\mathcal{N}_s \times \mathcal{N}_t))$ for all $i < 2$, $s \in \kappa^{<\mathbb{N}}$ and $t \in \mathbb{N}^{<\mathbb{N}}$.

Following the usual convention, we identify $\bigcup_{n \in \mathbb{N}} \kappa^n \times \mathbb{N}^n \times \mathbb{N}^n$ with $(\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$. Set $T = \{(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}} \mid A_{0,s,t} \cap \overline{A_{1,s,u}} \neq \emptyset\}$.

Lemma 3.2. *The tree T is well-founded.*

Proof. Suppose, towards a contradiction, that $(\sigma, c_0, c_1) \in [T]$. Then $(\sigma, c_i) \in C_i$ for all $i < 2$. Set $x_i = (\text{proj}_X \circ \phi_i)(\sigma, c_i)$ for all $i < 2$. As A_0 and A_1 are disjoint, it follows that $x_0 \neq x_1$, so there are disjoint open neighborhoods U_0 and U_1 of x_0 and x_1 . Fix $n \in \mathbb{N}$ such that $A_{i, \sigma \upharpoonright n, c_i \upharpoonright n} \subseteq \kappa^{\mathbb{N}} \times U_i$ for all $i < 2$. Then $A_{0, \sigma \upharpoonright n, c_0 \upharpoonright n} \cap \overline{A_{1, \sigma \upharpoonright n, c_1 \upharpoonright n}} = \emptyset$, which contradicts the fact that $(\sigma \upharpoonright n, c_0 \upharpoonright n, c_1 \upharpoonright n) \in T$. $\square$

Recursively define $B_{s,t,u} \subseteq \kappa^{\mathbb{N}} \times X$ for $(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$ by

$$B_{s,t,u} = \begin{cases} \nabla_{r \in S_s} \bigcup_{m \in \mathbb{N}} \bigcap_{n \in \mathbb{N}} B_{r, t \smallfrown (m), u \smallfrown (n)} & \text{if } (s, t, u) \in T \text{ and} \\ \sim A_{1,s,u} & \text{otherwise,} \end{cases}$$

where $S_s = \{s \smallfrown (\alpha) \mid \alpha < \kappa\}$. The usual induction shows that $B_{s,t,u}$ has uniformly Borel sections and separates $A_{0,s,t}$ from $A_{1,s,u}$, so $B_{\emptyset, \emptyset, \emptyset}$ separates A_0 from A_1 . $\square$

Proposition 3.3. *Suppose that κ is an aleph, X is a Hausdorff space, $A_0, A_1 \subseteq \kappa^{\mathbb{N}} \times X$ have uniformly analytic sections, and there are comeagerly many $\sigma \in \kappa^{\mathbb{N}}$ for which $(A_0)_\sigma \cap (A_1)_\sigma = \emptyset$. Then comeagerly many of their sections are separated by a set with uniformly Borel sections.*

Proof. Fix a dense G_δ set $C \subseteq \kappa^{\mathbb{N}}$ with $(A_0)_\sigma \cap (A_1)_\sigma = \emptyset$ for all $\sigma \in C$. As Propositions 1.4 and 1.6 ensure that the set $A'_0 = A_0 \cap (C \times X)$ has uniformly analytic sections, an application of Proposition 3.1 to A'_0 and A_1 yields the desired set. $\square$

Proposition 3.4. *Suppose that κ is an aleph, X_0 and X_1 are Hausdorff spaces, $A_i \subseteq \kappa^{\mathbb{N}} \times X_i$ for all $i < 2$ and $R \subseteq \kappa^{\mathbb{N}} \times (X_0 \times X_1)$ have uniformly analytic sections, and $((A_0)_\sigma \times (A_1)_\sigma) \cap R_\sigma = \emptyset$ for comeagerly many $\sigma \in \kappa^{\mathbb{N}}$. Then there are sets $B_i \subseteq \kappa^{\mathbb{N}} \times X_i$ for all $i < 2$ with uniformly Borel sections such that $(A_0)_\sigma \subseteq (B_0)_\sigma$, $(A_1)_\sigma \subseteq (B_1)_\sigma$, and $((B_0)_\sigma \times (B_1)_\sigma) \cap R_\sigma = \emptyset$ for comeagerly many $\sigma \in \kappa^{\mathbb{N}}$.*

Proof. The set $A'_0 = \text{proj}_{\kappa^{\mathbb{N}} \times X_0}((\text{proj}_{\kappa^{\mathbb{N}} \times X_1} \upharpoonright R)^{-1}(A_1))$ has uniformly analytic sections by two applications of Proposition 1.2. As comeagerly many of these sections are disjoint from those of A_0 , Proposition 3.3 yields a set $B_0 \subseteq \kappa^{\mathbb{N}} \times X_0$ with uniformly Borel sections separating comeagerly many sections of A_0 from those of A'_0 .

Along similar lines, the set $A'_1 = \text{proj}_{\kappa^{\mathbb{N}} \times X_1}((\text{proj}_{\kappa^{\mathbb{N}} \times X_0} \upharpoonright R)^{-1}(B_0))$ has uniformly analytic sections by Propositions 1.2, 1.5, and 1.6. As comeagerly many of these sections are disjoint from those of A_1 , Proposition 3.3 yields a set $B_1 \subseteq \kappa^{\mathbb{N}} \times X_1$ with uniformly Borel sections separating comeagerly many sections of A_1 from those of A'_1 . $\square$

Proposition 3.5. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ have uniformly analytic sections,*

and A_σ is G_σ -independent for comeagerly many $\sigma \in \kappa^\mathbb{N}$. Then there is a set $B \subseteq \kappa^\mathbb{N} \times X$ with uniformly Borel sections such that B_σ is a G_σ -independent superset of A_σ for comeagerly many $\sigma \in \kappa^\mathbb{N}$.

Proof. By Proposition 3.4, there exist $B_0, B_1 \subseteq \kappa^\mathbb{N} \times X$ with uniformly Borel sections, $A_\sigma \subseteq (B_i)_\sigma$ for all $i < 2$, and $((B_0)_\sigma \times (B_1)_\sigma) \cap G_\sigma = \emptyset$ for comeagerly many σ . Then $B_0 \cap B_1$ is as desired. $\square$

Identifying $\kappa^\mathbb{N}$ with $\mathcal{N}_t$ via the map $\sigma \mapsto t \frown \sigma$, these results have corresponding versions with the parameter space restricted to any basic open set $\mathcal{N}_t$. We use these versions without further comment.

4. THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

We use the notation and basic facts concerning $\mathbb{G}_0$, colorings, and homomorphisms from [dRM21, §1]. In particular, we use without further mention the fact that every $\mathbb{G}_0$ -independent set with the Baire property is meager. Let $s_n \in 2^n$ denote the sequences defining $\mathbb{G}_0$.

Proof of Theorem 1. Define $p_i: \kappa^\mathbb{N} \times (X \times X) \rightarrow \kappa^\mathbb{N} \times X$ by setting $p_i(\sigma, (x_0, x_1)) = (\sigma, x_i)$ for all $i < 2$, $\sigma \in \kappa^\mathbb{N}$, and $x_0, x_1 \in X$. As Propositions 1.1 and 1.2 ensure that $\bigcup_{i < 2} p_i(G)$ has uniformly analytic sections, there are closed sets $C, D \subseteq \kappa^\mathbb{N} \times \mathbb{N}^\mathbb{N}$ and continuous parameter-preserving functions $\phi: C \rightarrow \kappa^\mathbb{N} \times X$ and $\psi: D \rightarrow \kappa^\mathbb{N} \times (X \times X)$ such that $\bigcup_{i < 2} p_i(G) = \phi(C)$ and $G = \psi(D)$.

As in the classical proof, we will recursively construct a decreasing sequence $(B^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^\mathbb{N} \times X$ with uniformly Borel sections, beginning with $B^0 = \kappa^\mathbb{N} \times X$. At each stage we will remove sets with uniformly Borel independent sections associated with terminal approximations. As terminality might only hold locally in the parameter space, these sets will also be indexed by members of $\kappa^{<\mathbb{N}}$.

An *approximation* is a triple $a = (n^a, \phi^a, (\psi_k^a)_{k < n^a})$, where $n^a \in \mathbb{N}$, $\phi^a: 2^{n^a} \rightarrow \mathbb{N}^{n^a}$, and $\psi_k^a: 2^{n^a - (k+1)} \rightarrow \mathbb{N}^{n^a}$ for all $k < n^a$. An approximation b is a *one-step extension* of a if:

- (1) $n^b = n^a + 1$,
- (2) $\forall u \in 2^{n^a} \forall v \in 2^{n^b} (u \sqsubseteq v \implies \phi^a(u) \sqsubseteq \phi^b(v))$, and
- (3) $\forall k < n^a \forall u \in 2^{n^a - (k+1)} \forall v \in 2^{n^b - (k+1)} (u \sqsubseteq v \implies \psi_k^a(u) \sqsubseteq \psi_k^b(v))$.

For all $\sigma \in \kappa^\mathbb{N}$, a σ -*configuration* is a triple $\gamma = (n^\gamma, \phi^\gamma, (\psi_k^\gamma)_{k < n^\gamma})$, where $\phi^\gamma: 2^{n^\gamma} \rightarrow \mathbb{N}^{n^\gamma}$ and $\psi_k^\gamma: 2^{n^\gamma - (k+1)} \rightarrow \mathbb{N}^{n^\gamma}$ for all $k < n^\gamma$, such that:

- (1) $\forall u \in 2^{n^\gamma} \phi^\gamma(u) \in C_\sigma$,
- (2) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)} \psi_k^\gamma(v) \in D_\sigma$, and
- (3) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)}$
 $(\text{proj}_{X \times X} \circ \psi_\sigma \circ \psi_k^\gamma)(v) = ((\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_k \frown (i) \frown v))_{i < 2}$.

We say that γ is *compatible* with $Y \subseteq X$ if $(\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(2^{n^\gamma}) \subseteq Y$, and *compatible* with a if:

- (1) $n^\gamma = n^a$,
- (2) $\forall u \in 2^{n^a} \phi^a(u) \sqsubseteq \phi^\gamma(u)$, and
- (3) $\forall k < n^a \forall v \in 2^{n^a - (k+1)} \psi_k^a(v) \sqsubseteq \psi_k^\gamma(v)$.

Given a set $Y \subseteq \kappa^\mathbb{N} \times X$, we say that a is Y_σ -*terminal* if no σ -configuration is compatible with Y_σ and a one-step extension of a . Let $A(a, Y)$ denote the set of pairs of the form $(\phi_\sigma \circ \phi^\gamma)(\mathfrak{s}_{n^a})$, where $\sigma \in \kappa^\mathbb{N}$ and γ is a σ -configuration compatible with a and Y_σ .

The following fact can be proven in the exact same way as in the classical proof of the $\mathbb{G}_0$ dichotomy (see [dRM21, Lemma 1.5]):

Lemma 4.1. *Suppose that a is an approximation, $\sigma \in \kappa^\mathbb{N}$, $Y \subseteq \kappa^\mathbb{N} \times X$, and a is Y_σ -terminal. Then $A(a, Y)_\sigma$ is G_σ -independent.*

From this point forward, however, our proof will be a bit different.

Lemma 4.2. *Suppose that a is an approximation and $B \subseteq \kappa^\mathbb{N} \times X$ has uniformly Borel sections. Then $A(a, B)$ has uniformly analytic sections and $\{\sigma \in \kappa^\mathbb{N} \mid a \text{ is not } B_\sigma\text{-terminal}\}$ has the Baire property.*

Proof. The set of pairs (σ, γ) for which γ is a σ -configuration compatible with a is closed, and therefore has uniformly Borel sections by Proposition 1.4, so the set of such pairs for which γ is also compatible with B_σ has uniformly Borel sections by Proposition 1.5, and therefore uniformly analytic sections by Proposition 1.6, thus $A(a, B)$ has uniformly analytic sections by Proposition 1.2. It only remains to note that if A' is the set of all one-step extensions of a , then $\{\sigma \in \kappa^\mathbb{N} \mid a \text{ is not } B_\sigma\text{-terminal}\} = \bigcup_{a' \in A'} \text{proj}_{\kappa^\mathbb{N}}(A(a', B))$, which has the Baire property by Proposition 2.2. $\square$

Let $\mathcal{A}$ denote the countable set of approximations. Suppose that $B \subseteq \kappa^\mathbb{N} \times X$ has uniformly Borel sections, $s \in \kappa^{<\mathbb{N}}$, and $a \in \mathcal{A}$ is B_σ -terminal for comeagerly many $\sigma \in \mathcal{N}_s$. Lemma 4.1 and Proposition 3.5 yield a set $B_s(a, B) \subseteq \kappa^\mathbb{N} \times X$ with uniformly Borel sections such that $\forall^* \sigma \in \mathcal{N}_s$ $B_s(a, B)_\sigma$ is a G_σ -independent superset of $A(a, B)_\sigma$. We can clearly assume that $B_s(a, B)_\sigma = \emptyset$ whenever $\sigma \notin \mathcal{N}_s$.

Unlike the classical proof, we will obtain $(B^\alpha)_{\alpha < \kappa^+}$ from an increasing sequence $(S^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^{<\mathbb{N}} \times \mathcal{A}$, beginning with $S^0 = \emptyset$ and setting $S^\delta = \bigcup_{\alpha < \delta} S^\alpha$ for all limit ordinals $\delta < \kappa^+$. Given $\alpha < \kappa^+$ for which we have already found $S^\alpha \subseteq \kappa^{<\mathbb{N}} \times \mathcal{A}$, define $\rho^\alpha: S^\alpha \rightarrow \alpha$ by setting $\rho^\alpha(s, a) = \min\{\beta < \alpha \mid (s, a) \in S^{\beta+1}\}$ and $B^\alpha = \sim \nabla_{s \in \kappa^{<\mathbb{N}}} \bigcup_{(s, a) \in S^\alpha} B_s(a, B^{\rho^\alpha(s, a)})$, in addition to setting $S^{\alpha+1} = S^\alpha \cup \{(s, a) \in \sim S^\alpha \mid \forall^* \sigma \in \mathcal{N}_s \text{ } a \text{ is } B_\sigma^\alpha\text{-terminal}\}$, and fix $B_s(a, B^\alpha)$ as

above for all $(s, a) \in S^{\alpha+1} \setminus S^\alpha$. As $|\kappa^{<\mathbb{N}} \times \mathcal{A}| = \kappa$, there is a least $\lambda < \kappa^+$ for which $S^\lambda = S^{\lambda+1}$. Set $B = B^\lambda$, $\rho = \rho^{\lambda+1}$, and $S = S^\lambda$.

Given an approximation a and a one-step extension b of a , let $E(a, b)$ be the set of $\sigma \in \kappa^\mathbb{N}$ for which b is not B_σ -terminal and there is a σ -configuration compatible with b and B_σ . Lemma 4.2 ensures that these sets have the Baire property.

Lemma 4.3. *Suppose that $t \in \kappa^{<\mathbb{N}}$, a is an approximation such that $\forall^* \sigma \in \mathcal{N}_t$ a is not B_σ -terminal, and $\mathcal{B}$ is the set of one-step extensions of a . Then $\bigcup \{E(a, b) \mid b \in \mathcal{B}\}$ is comeager in $\mathcal{N}_t$.*

Proof. For all $b \in \mathcal{B}$, let $F(a, b)$ denote the set of $\sigma \in \kappa^\mathbb{N}$ for which there is a σ -configuration compatible with b and B_σ . As the set of $\sigma \in \mathcal{N}_t$ for which a is not B_σ -terminal is the union of these sets, it is enough to show that $F(a, b) \setminus E(a, b)$ is meager.

Suppose, towards a contradiction, that there exists $s \in \kappa^{<\mathbb{N}}$ with the property that b is B_σ -terminal and there is a σ -configuration compatible with b and B_σ for comeagerly many $\sigma \in \mathcal{N}_s$. Then $(s, b) \in S$, so $(\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_{n^b}) \in B_s(b, B)_\sigma$ for comeagerly many $\sigma \in \mathcal{N}_s$ and every σ -configuration γ compatible with b and B_σ . It then follows that $\forall \sigma \in \mathcal{N}_s$ $B_\sigma \cap B_s(b, B^{\rho(s, b)})_\sigma = \emptyset$, contradicting the fact that γ is compatible with B_σ . $\square$

Let a_0 denote the unique approximation for which $n^{a_0} = 0$.

Lemma 4.4. *Suppose that $t \in \kappa^{<\mathbb{N}}$ and a_0 is not B_σ -terminal for comeagerly many $\sigma \in \mathcal{N}_t$. Then there exist a dense G_δ set $H \subseteq \mathcal{N}_t$ and a continuous function $\pi: H \times 2^\mathbb{N} \rightarrow X$ such that π_σ is a homomorphism from $\mathbb{G}_0$ to G_σ for all $\sigma \in H$.*

Proof. Set $S_0 = \{t\}$ and $a_t = a_0$. Given $n \in \mathbb{N}$, an antichain $S_n \subseteq \kappa^{<\mathbb{N}}$, and approximations a_s for each $s \in S_n$, associate with each $r \in S_n$ the set $S_{n+1, r}$ of all minimal proper extensions s of r for which there is a one-step extension a_s of a_r such that $E(a_r, a_s)$ is comeager in $\mathcal{N}_s$. Lemma 4.3 ensures that $S_{n+1, r}$ is dense below r . Set $S_{n+1} = \bigcup_{r \in S_n} S_{n+1, r}$.

For all $n \in \mathbb{N}$, set $U_n = \bigcup_{s \in S_n} \mathcal{N}_s$. For all $\sigma \in U_n$, let $\ell_n(\sigma)$ be the unique natural number for which $\sigma \upharpoonright \ell_n(\sigma) \in S_n$. Fix a dense G_δ set $H \subseteq \bigcap_{n \in \mathbb{N}} U_n$ such that $\forall \sigma \in H \forall n \in \mathbb{N}$ $\sigma \in E(a_{\sigma \upharpoonright \ell_n(\sigma)}, a_{\sigma \upharpoonright \ell_{n+1}(\sigma)})$.

Define $\phi', \psi'_k: H \times 2^\mathbb{N} \rightarrow \mathbb{N}^\mathbb{N}$ by setting $\phi'(\sigma, c) = \bigcup_{n \in \mathbb{N}} \phi^{a_{\sigma \upharpoonright \ell_n(\sigma)}}(c \upharpoonright n)$ and $\psi'_k(\sigma, c) = \bigcup_{n > k} \psi_k^{a_{\sigma \upharpoonright \ell_n(\sigma)}}(c \upharpoonright n - (k+1))$ for all $c \in 2^\mathbb{N}$, $k \in \mathbb{N}$, and $\sigma \in H$. The existence of compatible configurations ensures that $(\sigma, \phi'(\sigma, c)) \in C$ and $(\sigma, \psi'_k(\sigma, c)) \in D$.

Define $\pi: H \times 2^\mathbb{N} \rightarrow X$ by setting $\pi(\sigma, c) = (\text{proj}_X \circ \phi_\sigma)(\phi'(\sigma, c))$ for all $c \in 2^\mathbb{N}$ and $\sigma \in H$. To see that π is the desired homomorphism,

observe that $(\text{proj}_{X \times X} \circ \psi_\sigma)(\psi'_k(\sigma, c)) = (\pi(\sigma, s_k \smallfrown (i) \smallfrown c))_{i < 2}$ for all $\sigma \in H$, $k \in \mathbb{N}$, and $c \in 2^\mathbb{N}$. $\square$

By Lemma 4.2, the set of parameters for which a_0 is not B_σ -terminal has the Baire property. If it is non-meager, then it is comeager in some $\mathcal{N}_t$, in which case Lemma 4.4 yields (2). If it is meager, then $(\emptyset, a_0) \in \text{dom}(\rho)$. Set $\alpha = \rho(\emptyset, a_0)$. As $A_\sigma(a_0, B) = B_\sigma \cap \bigcup_{i < 2} p_i(G)_\sigma$ for all $\sigma \in \kappa^\mathbb{N}$ and $B \subseteq B^\alpha$, it follows that $A_\sigma(a_0, B) = \emptyset$ for comeagerly many σ , thus B_σ is G_σ -independent for comeagerly many σ . For all $a \in \mathcal{A}$ and $n \in \mathbb{N}$, define $S_{a,n} = \{s \in \kappa^\mathbb{N} \mid (s, a) \in \text{dom}(\rho)\}$ and $B_{a,n} = \bigcap_{s \in S_{a,n}} B_s(a, B^{\rho(s,a)})$. Then $(B_{a,n})_\sigma$ is G_σ -independent for comeagerly many σ . The fact that $X = B_\sigma \cup \bigcup_{a \in \mathcal{A}, n \in \mathbb{N}} (B_{a,n})_\sigma$ for all $\sigma \in \kappa^\mathbb{N}$ therefore yields (1).

To see that (1) and (2) are mutually exclusive, simply observe that if both hold, then there are comeagerly many $\sigma \in \mathcal{N}_t$ for which $c_\sigma \circ \pi_\sigma$ is a Borel $\mathbb{N}$ -coloring of $\mathbb{G}_0$, a contradiction. $\square$

5. STRATIFICATIONS AND CATEGORY TRANSFORMS

Given a set $B \subseteq \kappa^\mathbb{N} \times X$ with uniformly Borel sections and $t \in \kappa^{<\mathbb{N}}$, define $B^{*t} = \{x \in X \mid B^x \text{ is comeager in } \mathcal{N}_t\}$.

Proposition 5.1. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, $B \subseteq \kappa^\mathbb{N} \times X$ and $B_s \subseteq \kappa^\mathbb{N} \times X$ have uniformly Borel sections for all $s \in \kappa^{<\mathbb{N}}$, and $U \subseteq X$. Then:*

- (1) $(\kappa^\mathbb{N} \times U)^{*t} = U$.
- (2) $\sim B^{*t} = \sim \bigcup_{u \supseteq t} B^{*u}$.
- (3) $(\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t} = \bigcap_{u \supseteq t} \bigcup_{v \supseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$.

Proof. As (1) and (2) are straightforward, we need only establish (3). Towards this end, note that if $x \in (\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is comeager in $\mathcal{N}_t$, so $\forall u \supseteq t \exists n \in \mathbb{N} \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, thus $\forall u \supseteq t \exists s \in \kappa^{<\mathbb{N}} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, and it follows that $\forall u \supseteq t \exists v \supseteq u \exists n \leq |v| B_{v \upharpoonright n}^x$ is comeager in $\mathcal{N}_v$, hence $x \in \bigcap_{u \supseteq t} \bigcup_{v \supseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$. And if $x \notin (\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is not comeager in $\mathcal{N}_t$, so there exists $u \supseteq t$ such that $(\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is meager in $\mathcal{N}_u$, thus $\forall v \supseteq u \forall n \leq |v| B_{v \upharpoonright n}^x$ is meager in $\mathcal{N}_v$, hence $x \notin \bigcup_{v \supseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$. $\square$

A subset of a topological space is κ^+ -Borel if it belongs to the smallest family containing the open sets and closed under complements and unions of cardinality at most κ .

Proposition 5.2. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then B^{*t} is κ^+ -Borel.*

Proof. By the obvious induction using Proposition 5.1. $\square$

A set $A \subseteq X$ is κ -Souslin if there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$. We say that a set $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a *stratification* of a κ -Souslin set A if there are witnesses $C \subseteq \kappa^{\mathbb{N}}$ and $\phi: C \rightarrow X$ to the κ -Souslinity of A with the property that $A' = \phi'(C')$, where $C' = \{(\sigma, b) \in \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \mid \sigma \circ b \in C\}$ and $\phi': C' \rightarrow \kappa^{\mathbb{N}} \times X$ is given by $\phi'(\sigma, b) = (\sigma, \phi(\sigma \circ b))$. For each $\sigma \in \kappa^{\mathbb{N}}$, we view A'_σ as the *analytic trace* of A on the set $\sigma(\mathbb{N})$.

Proposition 5.3. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq X$ is κ -Souslin, $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a stratification of A , and $x \in A$. Then $\forall^* \sigma \in \kappa^{\mathbb{N}}$ $x \in A'_\sigma$.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$ and $A' = \phi'(C')$, as well as $\tau \in C$ for which $\phi(\tau) = x$. Then the set $H = \bigcap_{n \in \mathbb{N}} \bigcup_{s \in \kappa^{<\mathbb{N}}} \mathcal{N}_{s \frown (\tau(n))}$ is a dense G_δ subset of $\kappa^{\mathbb{N}}$ and $\forall \sigma \in H$ $x \in A'_\sigma$. $\square$

Proposition 5.4. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a Hausdorff space, G is a κ -Souslin digraph on X , G' is a stratification of G , $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections, and B^{*t} is not G -independent. Then $\forall^* \sigma \in \mathcal{N}_t$ B_σ is not G'_σ -independent.*

Proof. Fix $x, y \in B^{*t}$ for which $x G y$ and appeal to Proposition 5.3 to see that $\forall^* \sigma \in \mathcal{N}_t$ $(x, y \in B_\sigma \text{ and } x G'_\sigma y)$. $\square$

We can now recover Kanovei's generalization of the $\mathbb{G}_0$ dichotomy.

Theorem 5.5 (Kanovei). *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and G is a κ -Souslin digraph on X . Then at least one of the following holds:*

- (1) *There is a κ^+ -Borel coloring $c: X \rightarrow \kappa$ of G .*
- (2) *There is a continuous homomorphism from $\mathbb{G}_0$ to G .*

Proof. Fix a stratification G' of G . As every homomorphism from $\mathbb{G}_0$ to a vertical section of G' is also a homomorphism from $\mathbb{G}_0$ to G , Theorem 1 allows us to assume that there is a uniformly Borel function $c': \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ such that $\forall^* \sigma \in \kappa^{\mathbb{N}}$ c'_σ is a coloring of G'_σ . For all $n \in \mathbb{N}$, set $B_n = (c')^{-1}(\{n\})$. For all $t \in \kappa^{<\mathbb{N}}$, Proposition 5.2 ensures that B_n^{*t} is κ^+ -Borel and Proposition 5.4 implies that it is G -independent. To see that the desired coloring can be obtained by disjointifying these sets,

it only remains to show that they cover X . Towards this end, suppose that $x \in X$ and observe that $\bigcup_{n \in \mathbb{N}} B_n^x$ is comeager. Then there exists $n \in \mathbb{N}$ for which B_n^x is non-meager, and therefore comeager in $\mathcal{N}_t$ for some $t \in \kappa^{<\mathbb{N}}$, so $x \in B_n^{*t}$. $\square$

Acknowledgements. ChatGPT provided substantial assistance in the discovery and development of the results.

REFERENCES

- [dRM21] N. de Rancourt and B. D. Miller, *The Feldman–Moore, Glimm–Effros, and Lusin–Novikov theorems over quotients*, 2021.
- [Kan97] Vladimir Kanovei, *Two dichotomy theorems on colourability of non-analytic graphs*, *Fundamenta Mathematicae* **154** (1997), no. 2, 183–201.
- [Kec95] Alexander S. Kechris, *Classical descriptive set theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.